

XOR PROPERTIES

```
// a ^ a = 0          (anything XORed with itself is 0)
// a ^ 0 = a          (XOR with 0 changes nothing)
// XOR is commutative and associative – order doesn't matter
x ^= y; // toggles bits of x wherever y has a 1
Demo: find the single number that appears once (others appear twice)
int result=0; for(int x:a) result^=x; // pairs cancel via a^a=0
Demo: find the missing number in array of 0..n (one missing)
int missing=0;
for (int i=0;i<=n;i++) missing^=i; // XOR all numbers 0..n
for (int x:a) missing^=x; // XOR all array elements
// remaining value is the missing number
```

BIT MANIPULATION (basic)

```
(n >> i) & 1; // check if bit i is set
n | (1 << i); // set bit i to 1
n & ~(1 << i); // clear bit i (set to 0)
n ^ (1 << i); // toggle bit i
n & (n - 1); // removes the lowest set bit
n & (-n); // isolates the lowest set bit (lowbit)
n << 1; n >> 1; // multiply/divide by 2 (fast)
Demo: check if a number is a power of 2
bool isPowerOf2 = (n > 0) && ((n & (n-1)) == 0);
Demo: generate all subsets of a set using bitmasks (n up to ~20)
for (int mask=0; mask<(1<<n); mask++) {
    for (int i=0;i<n;i++) {
        if (mask & (1<<i)) { /* element i is in this subset */ }
    }
}
```

builtin popcount

```
__builtin_popcount(x); // number of set bits (int)
__builtin_popcountll(x); // same, for long long
__builtin_clz(x); // count leading zeros (int)
__builtin_ctz(x); // count trailing zeros – position of lowest set bit
Demo: count total set bits across an array
long long totalBits=0; for(int x:a) totalBits += __builtin_popcount(x);
```

===== BONUS: TWO numbers that appear once (all others appear twice) =====

```
// Demo: find both unique numbers using XOR + lowbit isolation
int xorAll = 0;
for (int x : a) xorAll ^= x; // xorAll = num1 ^ num2

int diffBit = xorAll & (-xorAll); // isolate one bit where num1 and num2 differ

int num1 = 0, num2 = 0;
for (int x : a) {
    if (x & diffBit) num1 ^= x; // group 1: bit is set
    else num2 ^= x; // group 2: bit is not set
}
// num1 and num2 are the two unique numbers
```